Display Reversion Control Panel – General data

1. General

The Display Reversion Control Panel controls display brightness, power, and AGM reversion. Refer to Figure 1.



Display Reversion Control Panel
Figure 1

The display system allows the pilot to revert the Display Units (DU) and Advanced Graphics Module 1 and 2 (AGM) in the event of failure of an individual display unit or an AGM.

2. Description

A. Display Reversion Control Panel

The Display Reversion Control panel has potentiometers for the PILOTS PFD, UPPER MFD, LOWER MFD and CO-PILOTS PFD. The four potentiometers are used to adjust the individual Display Unit (DU) brightness and, at their most counterclockwise position, to switch the displays to OFF/REV.

The right side PFD has no Reversion (REV) function. Turning off the right side PFD simply turns off the display, and does not affect the contents of the other three display units. When the PILOTS PFD, UPPER MFD, or LOWER MFD potentiometers are moved to the OFF/REV position, the selected Display Unit (DU) goes blank and the information is moved to another display.

In some reversion cases (typically when two DUs have been turned off) the PFD will enter a composite mode. The PFD composite format shows the ADI/HSI, up to twelve CAS messages, the Systems Summary and the Radio windows, all on a single screen. Messages can be scrolled.

The two PFD controls also have a rotating switch surrounding their potentiometer that can be used to select the opposite side Advanced Graphics Module (AGM) in the event of a failure of the AGM normally assigned to drive that PFD. When the right hand rotating switch has been moved away from the NORM position and the Co-pilot PFD is being supplied with data from AGM 1, the engine instruments and radio window will appear on the outboard side of the Co-pilot's PFD.

The Display Reversion Control Panel is installed directly above the MF controller, on the aft face of the control column. Bezel and face lighting is controlled by the INSTRUMENT PANEL dim potentiometer.

3. Adjustment/Test

A. Function Test Display Reversion Control Panel

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Turn the PILOTS PFD dim control from the counterclockwise (MIN) position, to the fully clockwise position on the Reversion Control Panel. Verify that the Pilot's PFD display goes from dark to full brightness.

4) Turn the CO-PILOTS PFD dim control from the counterclockwise (MIN) position, to the fully clockwise position on the Reversion Control Panel. Verify that the Co-pilot's PFD display goes from dark to full brightness.

5) Turn the Multi-Function Displays (MFD) dim controls from the counterclockwise (MIN) position, to the fully clockwise position on the Reversion Control Panel. Verify that the Multi-Function Displays (MFD) display goes from dark to full brightness.

6) Set the PILOTS PFD, CO-PILOTS PFD, and Multi-Function Display dim controls from the full brightness position to a comfortable viewing dim level.

7) Set the PILOTS PFD revision control to AGM2. Verify the following:

   a) Pilot's PFD shows normal flight information.

   b) Co-pilot's PFD shows the pilot PFD format (radio window on the right side).

   c) Lower MFD shows a normal system multi-function display or situation awareness information with a Crew Alerting System (CAS) display in the upper left window.

   d) Upper MFD shows invalid (red cross).

8) Set the PILOTS PFD reversion control to NORM. Ensure the PFD and MFD displays return to normal.

9) Set the CO-PILOTS PFD revision control to AGM1. Verify the following:

   a) Co-pilot’s PFD shows normal flight information.

   b) Pilot’s PFD shows the co-pilot PFD format (radio window on the right side).

   c) Lower MFD shows invalid (red cross).

   d) Upper MFD shows a normal multi-function display with a Crew Alerting System (CAS) display in the upper left window.

10) Set the CO-PILOTS PFD reversion control to NORM. Ensure the PFD and MFD displays return to normal.

11) Set the PILOTS PFD dim control to OFF/REV. Verify that the Pilot's PFD is shown on the upper MFD.

12) Set the upper MFD dim control to OFF/REV. Verify that the Pilot's PFD is shown on the lower MFD.

13) Set the lower MFD dim control to OFF/REV. Verify that the composite PFD display is shown on the Co-pilot's PFD.

14) Set the CO-PILOTS PFD dim control to OFF. Verify that the composite PFD display is shown on the lower MFD.

15) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal and Installation

WARNING **INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:**

CHASSIS GROUNDING – PANEL / PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

A. Display Reversion Control Panel

1) Removal of Display Reversion Control Panel

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) To remove the Reversion Control Panel, turn the quick release fasteners.

c) Pull the Reversion Control Panel away from the controller mount pedestal to gain access to the electrical connector.

d) Disconnect electrical connector.

e) Remove unit from aircraft.

2) Installation of Display Reversion Control Panel

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Reconnect the electrical connector to the Reversion Control Panel.

c) Slide the Reversion Control Panel into the controller mount.

d) Turn the quick release fasteners to secure the Reversion Control Panel to the controller mount.

e) Do a function test, refer to Para 3.A.

5. Troubleshooting

A. Display Reversion Control Panel

1) Visually inspect the electrical connectors for damaged wiring or contacts. Ensure connectors are fully seated and tight.

2) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

3) If repair or service is required, send to appropriately rated avionics repair facility.

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